SYLLABUS OF COURSES

Parul® University

Subject Syllabus

303105453 - Cyber Physical Systems

Course: BTech **Semester:** 7

Prerequisite: Fundamentals of control systems,Basic knowledge of programming

Rationale : Cyber Physical Systems is a comprehensive understanding of the integration between computational algorithms and physical components in modern engineering systems. It emphasizes modeling, analysis, design, and implementation of CPS in real-world applications like autonomous vehicles, smart grids, and industrial automation.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	-	3	20	20	-	60	-	100

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	hisg	10	2
2	**Introduction to Internet of Things(IoT)** Embedded system – The core part of IoT, Application of Embedded Systems, Evolution in Computing, Machine to Machine(M2M) and IoT, What Is the Internet of Things (IoT)?, Components of IoT, Introduction to Cloud Computing, Introduction to Big data platforms for IoT, Introduction to Cloud and Fog ComputingEmbedded system – The core part of IoT, Application of Embedded Systems, Evolution in Computing, Machine to Machine(M2M) and IoT, What Is the Internet of Things (IoT)?, Components of IoT, Introduction to Cloud Computing, Introduction to Big data platforms for IoT, Introduction to Cloud and Fog Computing	10	3

3	**IoT Architecture** IoT Architecture layers, Three- and Five-Layer Architectures, Cloud and Fog Based Architectures **Breaking Down the IoT Stack:** **Devices** – Arduino, Raspberry Pi, NodeMCU(ESP8266) **Sensors & Actuators** - Buzzers, Relays, DC Motors, Stepper Motors, Servo Motors, Digital Sensors, Analog Sensors, Serial Communications with UART, Pulse width modulation, I2C, SPI	**20**	**8**
4	**IoT Protocol Stack** Zigbee, Zwave, RFID, NFC, Smart and Classic Bluetooth, Thread, CoAP, AMQP, DDS, MQTT, WiFi, LiFi , SMQTT, CoRE, 6LowPAN, 6TiSCH, RPL, CORPL, CARP	**20**	**9**
5	**Hardware and Software** **Arduino Board and C programming:** Arduino Platform, Arduino IDE, Compiling Code, Arduino Schematics, Arduino Basic Setup & Interface, Examples: Blink LED, Serial Print, ADC, pulse width modulation. **ESP8266 and Micropython/C:** ESP8266 platform, ESPlorer IDE, compiling code, Examples: Blink LED, Serial Print, ADC, pulse width modulation, Using ESP as Station, Uploading data on cloud, MQTT publish/subscribe, Communicating through android app **Raspberry Pi and Python:** About the board, Raspberry Pi Interfaces, Raspberry Pi vs. Arduino, Operating System Benefits, Raspberry Pi Setup, Introduction to Linux and Python, GPIO Access, Pulse width modulation, Blink LED, GUI using Tkinter, Network Programs, Client – server programs, Using Twitter API, Camera module, Servo control	**30**	**12**
6	**IoT Case Studies** IBM Watson, AmazonGo and SCiO. **Applications:** Efficient Waste Management in Smart Cities, A Smart Home Scenario, Shopping, Smart Healthcare systems, smart cities, IoT in industry.	**5**	**4**
7	**Open Challenges in IoT** Security, Scalability in Networking, Dynamic Topologies, Mobility, Reliability, Device Diversity and Interoperability, Integration of data from multiple Sources, Energy Efficiency, Bandwidth Management, Modeling and Analysis, Interfacing, Storage and computation to handle Exponential growth of data volume from Sensors, Complexity Management	**15**	**6**
	Total	**110**	**44**

Reference Books	
1.	**"Internet of Things (A Hands-on-Approach)",** By Vijay Madisetti and Arshdeep Bahga, | VPT
2.	**Designing the Internet of Things** By Adrian McEwen, Hakim Cassimall | Wiley | 1st
3.	**Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems** By Dr. Ovidiu Vermesan, Dr. Peter Friess | River Publishers
4.	**Interconnecting Smart Objects with IP: The Next Internet** By Jean-Philippe Vasseur, Adam Dunkels | Morgan Kuffmann Publishers
5.	**Getting Started with the Internet of Things** By Cuno Pfister, | O"Reilly Media